

General Overview and Institutional Profile

Institutional Identity and Status

The Balochistan University of Information Technology, Engineering and Management Sciences (BUITEMS) is an educational institution located in Quetta, Balochistan, Pakistan. The university is recognized as a world-class institution within the province, notably highlighted by the Express Tribune in May 2015.

BUITEMS holds international affiliations and certifications, including being an ISO 9001:2015 certified institution audited by TÜV Austria. It is an active member of six reputed international networks and associations, such as the International Association of Universities, The Talloires Network, and the American Society for Quality (ASQ). Additionally, the university serves as an official United Nations Academic Impact Hub for SDG 8 (Decent Work and Economic Growth) for the 2018–2021 tenure.

Strategic Metrics and Performance Indicators

Academic Rankings: Ranked number 4 by the Higher Education Commission (HEC) of Pakistan for Computer Science and Information Technology. Campus Vitality: Hosts more than 100 open days and events round the year. Employability: Achieves an 80% job placement ratio for graduates within their first year

of entering the workforce. Financial Access: One out of every three enrolled students receives financial assistance. Scale of Operations: Features a community of more than 8,000 students and a faculty body trained at leading global institutions. Strategic Governance and Leadership

Quality Policy Statement

BUITEMS operates under a formalized mandate to define standards and systems for the uplift of the socio-economic order through quality education and services. This mission is driven by several operational directives: Providing an environment highly conducive to learning, teaching, academic inquiry, and scientific innovation. Maintaining rigorous academic excellence and professional standards across all entities. Adhering strictly to established administrative systems to ensure robust corporate

governance for the management and transfer of knowledge. Benchmarking organizational performance against leading institutions of higher education to drive continuous institutional improvement. Enhancing the efficiency and effectiveness of university operations by actively encouraging the participation of stakeholders. Pursuing continuous development through corporate creativity, structured teamwork, and

adaptive change management. Playing a central catalytic role to achieve national, regional, and global harmony.

Administrative Structure

The operational workflow of the university is divided into six distinct faculties, each composed of multiple specialized academic departments. The Faculty and its constituent departments bear direct responsibility for:

Determining, designing, and continuously updating course curricula. Organizing and executing academic lectures, seminars, practical laboratory sessions, and research projects. Setting, administering, and

marking student examinations. Maintaining professional and academic excellence. Ensuring the preservation of an environment designed for educational inquiry. Central administrative support functions—including student enrollment, academic registries, and

quality control systems—are maintained globally through the office of the Registrar, the Controller of Examinations, and dedicated university Directorates. Executive Leadership Message

The Vice Chancellor, Professor Dr. Khalid Hafeez, emphasizes a strategic vision that blends

rigorous classroom learning with applied research, a real-world experiential focus, and a forward-looking "Future of Work" outlook. The university infrastructure includes high-tech laboratories equipped for industrial partnerships, along with specialized research, consulting services, and short courses designed for businesses and corporate entities. Academic operations run under the institutional guidance of the Honorable Governor of Balochistan, who serves as the university Chancellor. Executive Profile: Khalid Hafeez, PhD

Title: Vice Chancellor; Professor of AI and Digital Transformation. Professional Fellowships and Certifications: Fellow of the Royal Society of Arts (RSA-UK); Fellow of the Chartered Management Institute (CMI-UK); Senior Fellow of the Higher Education Academy (SFHEA-UK); Certified Six Sigma Master Black Belt (SSMBB-US); Certified European Foundation of Quality Management (EFQM-Brussels, Belgium); Certified Lean Systems and Process Manager (Herrington Institute, US);

Certified Project Manager (Herrington Institute-US); Certified Natural Linguistic Programming (NLP-UK).

Admissions Policy and Procedures

Candidate Profiles and Admission Criteria

BUITEMS targets self-motivated, disciplined individuals who exhibit critical thinking skills and possess strong academic backgrounds and potential. Candidates must demonstrate a clear

interest in their chosen discipline rather than just seeking a degree. The university selection mechanism measures academic capability via historical qualifications combined with standardized entry testing. Once admitted, students are expected to maintain high cumulative grade point averages (CGPA).

Step-by-Step Application Process

1. Major Selection: Candidates select a specific field of study (major) aligned with their long-term academic and professional interests. 2. Application Submission: Applicants complete the formal admission form via the online

portal (admissions.buitms.edu.pk) or physically at the campus admissions office. The form must be submitted along with requisite academic documentation and the official bank receipt for the admission processing fee. 3. Standardized Testing: Eligible applicants are issued an official test admit card and invited to take the National Testing Service (NTS) admission test, which follows standard protocols designed for the specified major. 4. Merit Evaluation and Admission Offer: Final admission offers are extended based on a

structured merit calculation derived from a combination of matriculation marks, intermediate marks, and individual performance on the NTS admission test. Rules for International and Transfer Students

The institution explicitly encourages applications from international students to foster

cross-border collaboration and campus diversity. The baseline academic eligibility criteria for international applicants match those applied to national students. For candidates seeking a transfer from external local or foreign higher education institutions, the applicant must have successfully passed the BUIITEMS admission test, securing a score equal to or higher than the minimum merit threshold of the target program. Final transfer approvals remain strictly subject to seat availability and a qualitative audit of the academic work completed at the previous institution.

Required Supporting Documentation

All applications must include the following verified documents: Attested copy of the Secondary School Certificate. Attested copy of the Higher Secondary School Certificate. Attested copy of the applicant's Computerized National Identity Card (CNIC) or B-form.

Attested copy of the applicant's Local/Domicile certificate. The original bank draft, pay order, or cash payment receipt for the processing fee. Six recent passport-sized photographs. Attested copy of the CNIC of the applicant's father or legal guardian. Attested copy of an official Character Certificate issued by the last educational institution attended.

Financial Transactions and Contact Channels

The mandatory, non-refundable application processing fee is set at Rs. 3,500 for national applicants (applying manually or online) and US \$75 for foreign applicants. Fees must be cleared on or before the specified application closing date using one of two approved payment options:

A formal bank draft or pay order drawn in favor of the Registrar BUIITEMS, Quetta. Direct cash deposition into Account Number 23587000000201 at Habib Bank Limited, BUIITEMS University Branch, Quetta, Pakistan. Hardcopy application packets and supporting credentials must be shipped directly to: Admissions Office, Balochistan University of Information Technology, Engineering and Management Sciences, Takatu Campus, Airport Road, Baleli, Quetta.

Universal Access Number (UAN): +92 81 111-717-111 Alternative Telephone Lines: +92 81 2880560, 2880136, 2880140, 2880386, 2880432, 2880527, 2881076, 2880511 Internal Extensions: 163, 216, 217
Academic Regulations and Policies

Program Durations and Structural Timelines

The standard duration for a Bachelor of Science (BS) degree program is four years, which is systematically divided into eight individual semesters. The Bachelor of Architecture (B. Arch) program is uniquely extended over five years, encompassing ten semesters. Each academic semester spans 18 weeks, providing 16 weeks of direct classroom instruction and 2 weeks for mid-term and final examinations. Master of Science (MS) programs are designed for a 2-year duration from the formal date of student enrollment.

Credit Hour Limits per Semester

Students must enroll in a standard course load ranging between 15 and 18 credit hours per regular semester. In exceptional circumstances, the Chairperson of the respective department holds the authority

to permit a maximum course load of up to 21 credit hours, subject to formal approval by the Dean of the faculty.

Academic Probation and Disqualification Rules

To successfully graduate, undergraduate students must secure a minimum final CGPA of 2.0, while MS scholars must achieve a minimum CGPA of 2.5. Academic standing is regulated by a strict three-tier probation system: First Probation: Triggered whenever a student's cumulative grade point average drops

within the critical window between 1.0 and 2.0. This status applies to the immediately following semester. Second Probation: Applied if the student fails to raise their CGPA to 2.0 or above by the conclusion of the first probationary semester. Last Probation: Applied if the student fails to remedy the academic deficit and raise their CGPA to at least 2.0 by the end of the second probationary semester. Academic Dismissal: If a student fails to achieve a minimum CGPA of 2.0 during their

last probation semester, they are dropped from the university rolls. Rules for Freezing Semesters and Program Changes

Freezing Policy: Students are prohibited from freezing their very first semester of a

program. After successfully completing the first semester, a student may apply to freeze a maximum of two semesters across their entire degree program. Program Transfer Policy: Alterations to a student's chosen academic program are generally discouraged. Transfers may be permitted within the first two weeks of the first semester, provided they are recommended by the relevant Department Chairpersons and approved by the Deans. The applicant must meet all base eligibility criteria, a seat must

be available, and the student's entry merit score must be higher than the merit score of the final student admitted into the target program during that intake cycle. Financial Assistance Framework

Financial aid options are administered through the University Advancement and Financial Assistance Office. Available support packages include:

Merit scholarships (accessible to students after completing their first semester). Dedicated Work and Study programs. Direct tuition fee concessions for financially needy students. Sibling fee concessions for families with multiple concurrent enrollments. Fee concessions for the direct dependents of active BUIITEMS employees. Staged payment plans allowing fee submission via installments.

The USAID Need-Based Scholarship program. Campus Life, Sports, and Infrastructure

Student Organizations and Culture

Student life at the university balances academic workloads with active co-curricular engagement, managed primarily by the Directorate of Student Affairs. The campus features more than 20 separate specialized student clubs and socio-cultural organizations, which

organize at least one campus event daily on average. Event scales range from small specialized gatherings of 50 people to massive flagship festivals drawing thousands of participants and external spectators. These major events include the Spring Festival, the Culture Feast, and Homecoming. The student body handles publishing tasks for a quarterly campus newsletter and the official university magazine, titled Markhor. Sports Infrastructure and the Annual Olympiad

The university maintains extensive sports infrastructure operated in conjunction with national bodies, tracking a registered user base of over 3,000 members. Over 1,000 students regularly participate in athletic tournaments inside and outside the province. The athletic system features several notable facilities: A newly constructed multi-purpose Sports Complex that provides professional-grade courts and tactical gear for basketball, handball, badminton, table tennis, and volleyball.

A high-class, fully equipped gym and dedicated fitness center situated directly inside the main Sports Complex. A professional cricket stadium built and maintained to Pakistan Cricket Board (PCB) standards, which regularly hosts official regional cricket tournaments.

A professional-grade, lush green football ground that operates as a core campus sporting hub throughout the year. The university athletic year culminates in the Annual Olympiad—a week-long tournament of non-stop sports and athletic competitions spanning over 15 distinct sports categories. The event involves over 1,000 student and staff athletes, drawing thousands of local spectators from across Quetta.

Campus Dining, Amenities, and Entertainment

The university features dedicated retail dining, social hubs, and food houses designed to facilitate intellectual debates, creative poetry recitations, and literary interactions. Active campus dining options include: Road Stoves

Hot & Chili Gosha-e-Fikr-o-Nosh Life Café City Café Civic Engagement and Community Service

BUITEMS students and staff maintain a strong commitment to civic involvement through self-formed public service programs. Key community initiatives include: Free Citizen Schools: An organization founded and run directly by BUITEMS personnel to provide elementary education to local child laborers. Volunteers teach children basic reading and writing skills in Urdu and English, alongside foundational concepts in mathematics and science, helping support their families' financial needs.

Weekend Community Outreach: Student teams visit specialized community schools in remote areas of Quetta on weekends to deliver modern educational instruction. Healthcare Support and Blood Drives: Regular blood donation campaigns that provide critical support to medical patients throughout the province. Career Counseling: Structured workshops and formal training sessions organized for young graduates from various regional universities to assist with career planning and

international scholarship applications. Geographical Context and Regional Tourism

The university is located in Quetta, known as the fruit basket of Pakistan, which is characterized by its high-altitude mountain backdrops, surrounding valleys, and fruit orchards. The regional environment features several prominent natural and cultural landmarks:

Ziarat Juniper Forest: Located approximately 125 kilometers north of Quetta, this site holds global distinction as the second-largest contiguous juniper forest in the world. Hanna Lake: Situated 10 kilometers east of the city, this turquoise water destination offers boat rentals and a lakeside restaurant for hikers and campers. Hazarganji Chiltan National Park: Located 20 kilometers southwest of Quetta, this protected park features the Chiltan Hill Viewpoint, which offers a panoramic views of the

city. Pir Ghaib: A natural waterfall and picnic spot located 70 kilometers from Quetta in the Bolan Valley. Kharkhasa: A narrow, 16-kilometer-long valley located 10 kilometers west of Quetta that contains diverse

flora and fauna species. Quetta Geological Museum: Located on Sariab Road, it houses an extensive collection

of regional rocks and historical fossils. Command and Staff College Museum: Dedicated to British military history and housed in the historic former bungalow of Field Marshal Bernard Montgomery. Quetta Archaeological Museum: Located on Wafa Road, it features displays of Stone Age tools, prehistoric pottery from Mehrgarh, rare antique firearms, historical swords, ancient manuscripts, and photographs documenting Quetta prior to the devastating 1935

earthquake. Balochistan Arts Council Library: Serves as a preservation repository for arts and crafts originating from the province. Urak Valley and Pishin: Local holiday stations recognized for fresh fruit production and natural forests. The campus coordinates guided university orientation tours for young schoolchildren from across the province. These visits expose children to science, engineering, and arts disciplines to

help them plan their future academic paths early in life. Faculty of Basic Sciences (FBS)

Institutional Genesis and Scope

The Faculty of Basic Sciences (FBS) was officially established in September 2020 (with fully realized operations by 2021) following a formal administrative bifurcation of the former Faculty of Arts and Basic Sciences (FABS). This split divided the original entity into two separate

faculties: the Faculty of Basic Sciences (FBS) and the Faculty of Social Sciences and Humanities (FSS&H). The Faculty of Basic Sciences serves as an essential academic foundation for the university, providing fundamental science coursework for competitive professional fields like medicine, engineering, and computer science. Because these core subjects underpin all applied sciences, FBS effectively outsources its faculty to deliver foundational coursework across nearly every department within BUIITEMS.

Academic Architecture and Degree Programs

The Faculty of Basic Sciences is organized into three distinct academic departments: Chemistry, Physics, and Mathematical Sciences. The individual degree pathways offered across these departments include:

Department of Chemistry: Offers a 4-year BS degree, a 2-year MS degree, and a 3-year PhD program.

Department of Physics: Offers a 4-year BS degree, a 2-year MS degree, and a 3-year PhD program.

Department of Mathematical Sciences: Offers a 4-year BS degree and a 2-year MS degree. (The formal documentation and approval process for its PhD program is currently in progress). Research Profile and Faculty Quality

The core operational focus of the Faculty of Basic Sciences is centered on high-impact teaching and academic research. The faculty members are highly qualified, with a significant majority having earned their doctoral degrees from leading international institutions. Faculty members across all three departments have secured competitive research grants from various national and international funding agencies to develop new scientific methodologies. Students within the faculty begin their degree programs by establishing a conceptual foundation across a broad spectrum of basic subjects before advancing to specialized, higher-level

coursework. Internal Faculty Governance

The internal management hierarchy of the Faculty of Basic Sciences is organized as follows: Dean: Prof. Dr. Hamdullah Khan Tareen

Associate Dean: Dr. Muhammad Nawaz Departmental Chairs: Chairperson, Department of Chemistry: Dr. Hameeda Panezai Chairperson, Department of Physics: Prof. Dr. H. M. Noor ul Huda Khan Asghar Chairperson, Department of Mathematical Sciences: Dr. Hasrat Hussain Shah Department of Chemistry

Overview and Research Focus

Established in 2015, the Department of Chemistry provides curriculum paths aligned with HEC regulations. Its academic roster includes 13 PhD-qualified faculty members and 4 MS-qualified lecturers, many of whom manage research projects valued in the millions. The department's research focus spans the synthesis, isolation, and characterization of natural products, alongside physical, polymer, and material chemistry. Faculty and students regularly publish their findings in peer-reviewed national and international scientific journals.

Strategic Vision, Mission, and Career Outcomes

Vision: To become a recognized model of chemical education that prepares students for the challenges of the 21st century, while fostering cross-disciplinary collaborations inside and outside the university.

Mission: To advance the physical sciences through research-centered learning, high

ethical standards, strong faculty-student interaction, and modern laboratory technology. Objectives: To achieve excellence in teaching and cutting-edge research; to equip the next generation of chemists to solve real-world problems; to drive regional economic transformation; and to cultivate advanced experimental skills. Career Scope: Graduates qualify for a wide range of professional roles, including

analytical chemists, physical chemists, laboratory chemists, production chemists, R&D managers, R&D directors, biomedical chemists, and industrial research scientists. Key employment sectors include pharmaceutical companies, agrochemical industries, glass, sugar, cement, ceramics, beverages, petrochemicals, plastic manufacturing, textiles, and the paper industry. Baseline Curriculum Architecture (BS Chemistry)

The curriculum follows the standardized format recommended by the HEC National Curriculum Revision Committees (NCRCs) for a 4-year integrated bachelor's degree: Compulsory Requirements (No Choice): 9 courses, totaling 25 credit hours (19.23% of the curriculum). General Courses (Chosen from other departments): 7 to 8 courses, totaling 21 to 24

credit hours (17.30% of the curriculum). Discipline-Specific Foundation Courses: 9 to 10 courses, totaling 30 to 33 credit hours (24.23% of the curriculum). Major Courses (Including research projects and internships): 11 to 13 courses, totaling 36 to 42 credit hours (30.00% of the curriculum). Elective Courses (Within the major discipline): 4 courses, totaling 12 credit hours

(9.23% of the curriculum). Total Program Requirements: 40 to 44 courses, accumulating between 124 and 136 total credit hours. Semester Constraints: 4 to 6 courses per semester, with a strict limit of no more than 3 concurrent laboratory/practical courses. BS Chemistry Semester-by-Semester Course Roadmap

First Semester: Inorganic Chemistry-I, Computer Fundamentals, Mathematics-I, Pakistan Studies, Reading and Writing Skills, Physics. Second Semester: Organic Chemistry-I, Environmental Chemistry,

Food and Nutrition Chemistry, Mathematics-II, Islamic Studies/Ethics, Academic Reading and Writing.
Third Semester: Physical Chemistry-I, Basic Forensic Chemistry, Applied Chemistry,

Introduction to Psychology, Statistics, Communication & Presentation Skills. Fourth Semester: Analytical Chemistry-I, Biochemistry-I, Soil Chemistry, Research Methods, Fuel Chemistry. Fifth Semester: Biochemistry-II, Inorganic Chemistry-II, Organic Chemistry-II, Physical Chemistry-II. Sixth Semester: Analytical Chemistry-II, Inorganic Chemistry-III, Organic Chemistry-III, Physical Chemistry-III.

Seventh Semester: Special Course-I, Special Course-II, Special Course-III, Special Course-IV, Special Course-V, Special Lab-I. Eighth Semester: Special Course-VI, Special Course-VII, Special Course-VIII, Special Lab-II, Research Project. Note on Specializations: Formal specialization tracks are offered during the seventh and eighth semesters, subject to faculty availability and student enrollment numbers. Five specialization

streams are available: Organic Chemistry, Inorganic Chemistry, Physical Chemistry, Analytical Chemistry, and Biochemistry. MS and PhD Chemistry Course Layouts

MS Chemistry (First Semester): Core Course-I, Core Course-II, Core Course-III,

Elective Course-I. MS Chemistry (Second Semester): Core Course-IV, Core Course-V, Elective Course-II, Elective Course-III. MS Chemistry (Third & Fourth Semesters): Independent Research Thesis. PhD Chemistry (First Semester): Elective I, Elective II, Elective III. PhD Chemistry (Second Semester): Elective IV, Elective V, Elective VI. PhD Chemistry (Third Semester onward): Doctoral Dissertation execution.

Pedagogical Constraint: PhD scholars are strictly barred from enrolling in courses they have already completed during their prior MS or M.Phil coursework. Available Laboratory Instrumentation

The Department of Chemistry maintains an inventory of specialized analytical instruments for

teaching and research: Fluorescence Spectrometer UV-Visible Spectrophotometer Flame Photometer Viscometer Stalagmometer Tensiometer

Digital pH Meter Digital Conductivity Meter Centrifuge Refractometer Fluorometer Microcentrifuge

Ultrasonic Cleanser Chemistry Faculty Profiles

Prof. M. Naeem Shahwani (Professor): PhD from the University of Glasgow, Scotland, UK. Research focus: Environmental stress on plant membranes using physiological, biochemical, and molecular/genetic approaches.

Dr. Muhammad Nawaz (Associate Professor & Associate Dean): PhD from the University of Rostock, Germany; Postdoctoral research at Korea University and the University of Sao Paulo, Brazil. Research focus: Organic synthesis, catalysis, and natural products chemistry. Dr. Hameeda Panezai (Professor & Chairperson): PhD from the Beijing University of Technology, China. HEC Approved Supervisor. Research focus: Synthesis of nano- and

microporous materials, surface property determination, and adsorption/absorption mechanisms of ordered porous materials for environmental remediation and energy storage. Dr. Fouzia Abdul Sattar (Professor): PhD from HEJRIC, University of Karachi, Pakistan; Postdoctoral research at the University of Karachi, University of Malaya (Malaysia), University of Bath (UK), Trinity College Dublin (Ireland), and FJIRSM

(China). Research

focus: Medicinal chemistry, catalysis, natural products, and organic synthesis. Dr. Fayaz Ahmed (Professor): PhD from HEJRIC, University of Karachi, Pakistan; Postdoctoral research at the University of Karachi and FJIRSM (China). Research focus: Natural products, synthetic methodologies, surface functionalization, and organic synthesis. Dr. Gulfam Nasar (Associate Professor): PhD from the University of Peshawar,

Pakistan. Research focus: Nanomaterials, polymer composites/nanocomposites, ferrites, electromagnetic materials, material science, and innovative construction materials. Dr. Shazia Iqbal (Associate Professor): PhD from HEJRIC, University of Karachi, Pakistan. Research focus: Natural product chemistry, environmental microbiology, and environmental chemistry. Dr. Mamoon Ur Rashid (Associate Professor): PhD from the University of Peshawar, Pakistan. Research focus: Medicinal chemistry and natural products.

Dr. Hamid Ullah (Associate Professor): PhD from the State University of Campinas, Brazil. HEC Approved Supervisor. Research focus: Organic synthesis and natural product chemistry. Dr. Aziza Sarwar (Assistant Professor): PhD from the University of Malaya, Malaysia. Research focus: Synthesis and characterization of azomethine Schiff bases, transition/lanthanide metal-azomethine complexes, luminescence studies, and

hydrothermal synthesis of ligands and complexes. Dr. Muhammad Raheel (Assistant Professor): PhD from Quaid-i-Azam University, Islamabad. Research focus: Physical chemistry, polymer nanocomposites, polymer carbonization, polymer foams, and surface/colloid chemistry (surfactant drug/DNA interactions). Dr. Mujtaba Ellahi (Assistant Professor): PhD from the University of Science and Technology, Beijing, China. HEC Approved Supervisor. Research focus: Nanochemistry

and nanotechnology. Dr. Zahoor Ullah (Assistant Professor): PhD from Universiti Teknologi PETRONAS, Malaysia. HEC Approved Supervisor. Research focus: Ionic liquids, biofuels, biomass, renewable energy, and natural products. Dr. Mohibullah (Assistant Professor): PhD from Quaid-i-Azam University, Islamabad. Research focus: Polymers, polymer composites, and the synthesis and application of

hybrid cryogels. Mr. Umair Ahmed (Lecturer - On Study Leave): MS and MSc from the University of Balochistan, Quetta. Research focus: Organic synthesis, drug design, molecular biology, genetic engineering, and protein synthesis. Samra Farooq (Lecturer): M.Phil. from Sardar Bahadur Khan (SBK) Women's University,

Quetta. Research focus: Organic synthesis, green synthesis, natural products, and nanochemistry. Ms. Noornamea (Lecturer): MS from BUIEMS, Quetta. Research focus: Biological and physicochemical analysis of water. Mr. Saiful Hanan (Lecturer): MS from BUIEMS, Quetta. Research focus: Inorganic complex analysis.

Department of Physics

Overview, Core Disciplines, and Infrastructure

The Physics curriculum was originally introduced in 2004 under the Faculty of Information and Communication Technology, with specialized tracks in BS Physics and MS Semiconductor Physics. The independent Department of Physics was formally established in 2007 under the former Faculty of Arts and

Basic Sciences. Following HEC clearance, the department added its

PhD program in 2017. The academic faculty includes eight PhD degree holders with international teaching and research experience, alongside eight faculty members currently on study leave. The department operates seven dedicated teaching laboratories for undergraduate and graduate studies, a Seminar Library housing books, research reports, monographs, and dissertations, and specialized computing infrastructure for mathematical modeling and graphical simulations. Advanced research is centered in two primary high-tech units: the Plasma Physics Laboratory

(PPL) and the Materials Research Laboratory (MRL). The department's specialized disciplines span: Materials science and energy materials Magnetic materials and spintronics Radiation and nuclear physics Plasma physics

Atomic and molecular physics Quantum optics, quantum mechanics, and optomechanics Theoretical high-energy physics and cosmology Future expansion plans include transforming the department into a dedicated School of Physics, introducing new research fields such as astronomy, medical physics, silicon technology, and theoretical materials physics.

Strategic Vision, Mission, and Learning Outcomes

Vision: To establish national leadership in pure and applied physics, match the academic credibility of technologically advanced countries, and cultivate competent physicists who excel at critical thinking and solving academic challenges. Mission: To provide high-quality education and cutting-edge research in physical science

and emerging technologies, inspiring students to pursue lifelong learning and explore the universe.

Objectives: To deliver quality education in physics; to advance research in emerging technologies; to emphasize experimental learning and innovative investigation techniques;

to train scholars for public and private scientific organizations; and to foster interdisciplinary collaborations both nationally and internationally. Outcomes: The department graduates physics professionals who secure roles in educational institutes and scientific organizations worldwide, with specialized training in plasma physics, solid-state physics, quantum optics, and radiation physics. BS Physics Semester-by-Semester Course Roadmap

First Semester: Mechanics, Lab-I, Calculus-I, Reading and Writing Skills, Pakistan Studies, Computer Fundamentals. Second Semester: Electricity and Magnetism-I, Heat and Thermodynamics, Lab-II, Islamic Studies/Ethics, Introduction to Geology, Calculus-II, Communication & Presentation Skills.

Third Semester: Electricity and Magnetism-II, Waves and Oscillations, Modern Physics, Lab-III, Linear Algebra, Academic Reading & Writing, Statistics-I. Fourth Semester: Optics, Lab-IV, Differential Equations, Introduction to Psychology, Statistics-II, Introduction to Chemistry, Programming Fundamentals. Fifth Semester: Mathematical Methods of Physics-I, Electromagnetic Theory-I, Classical Mechanics, Basic Electronic Devices-I, Statistical Physics, Lab-V.

Sixth Semester: Mathematical Methods of Physics-II, Quantum Mechanics-I, Electromagnetic Theory-II, Atomic & Molecular Physics, Basic Electronic Devices-II, Lab-VI. Seventh Semester: Quantum Mechanics-II, Solid State Physics-I, Lab-VII, Project Part-I, Elective I, Elective II. Eighth Semester: Solid State Physics-II, Nuclear Physics, Project Part-II, Elective III, Elective IV.

MS and PhD Physics Course Layouts

MS Physics (First Semester): Mathematical Methods of Physics, Electrodynamics, Methods and Techniques of Experimental Physics, Quantum Mechanics-I. MS Physics (Second Semester): Elective I, Elective II, Elective III, Elective IV.

MS Physics (Third Semester onward): Independent Research Thesis. PhD Physics (First Semester): Elective I, Elective II, Elective III. PhD Physics (Second Semester): Elective IV, Elective V, Elective VI. PhD Physics (Third Semester onward): Doctoral Dissertation execution. Pedagogical Constraint: PhD students are prohibited from retaking subjects they have already completed during their MS or M.Phil studies. Physics Faculty Profiles

Prof. Dr. Hamdullah Khan Tareen (Dean): PhD from Quaid-i-Azam University, Islamabad. HEC Approved Supervisor. Research focus: Plasma physics and material processing by plasma. Prof. Dr. H. M. Noor ul Huda Khan Asghar (Professor & Chairperson): PhD from

Dalian University of Technology, China. HEC Approved Supervisor. Research focus: Energy materials, magnetic materials, and nanotechnology. Prof. Dr. Syed Zaheer Abbas Gilani (Professor): PhD from The Islamia University of Bahawalpur, Pakistan. HEC Approved Supervisor. Research focus: Materials science, magnetic materials, and nanotechnology. Dr. Nisar Ahmad (Assistant Professor): PhD from Universiti Sains Malaysia. HEC

Approved Supervisor. Research focus: Soil physics and radiation physics. Dr. Muhammad Naeem Khan (Assistant Professor): PhD in Energy Studies (Applied Physics) from the University of Brunei Darussalam. HEC Approved Supervisor. Research focus: Perovskite solar cells and solid oxide fuel cells. Dr. Farooq (Assistant Professor): PhD from Dalian University of Technology, China. HEC Approved Supervisor. Research focus: Theoretical physics, quantum

optomechanics, quantum optics, and condensed matter physics. Dr. Rafaqat Hussain (Assistant Professor): PhD from COMSATS, Islamabad. Research focus: Spintronic materials, microfabrication, and nanospectroscopy. Muhammad Khalid (Assistant Professor): PhD (Final Stage) at Xi'an Jiaotong University, China. Research focus: Spintronic renewable energy, optical and electrical modeling, and photovoltaics. Dr. Sunila Bukhsh (Assistant Professor): PhD from USTC, China. Research focus:

Condensed matter physics and electronic structure calculation methods. Sumayya Abdul Wadood (Lecturer): MS from BUIITEMS, Pakistan. Research focus: Material science and quantum dots. Niamat Ullah Khan (Lecturer): PhD in progress at Quaid-i-Azam University, Islamabad. Research focus: Theoretical high-energy physics. Muhammad Akram Khan (Lecturer - On Study Leave): PhD in progress in the USA.

Research focus: Theoretical high-energy physics and cosmology. Dr. Javed Rehman (Lecturer): PhD from Jilin University, China. Research focus: Theoretical screening of 2D and 3D materials for metal-ion batteries. Sameen Aslam (Lecturer - On Study Leave): MS from BUIITEMS, Pakistan. Research focus: Nanotechnology and magnetic materials. Mr. Muhammad Saad (Lecturer - On Study Leave): MS from Quaid-i-Azam University, Islamabad. Research focus: Quantum magnetism, magnetic effects in 2D materials, and

transition metal oxides. Furhaj Ahmed Sheikh (Lecturer): MS from BUIITEMS, Pakistan. Research focus: Material science, nanotechnology, magnetic materials, and spinel-ferrites. Abdul Qayyum (Lecturer): MS

from BUIITEMS, Pakistan. Research focus: Quantum optics, quantum optomechanics, and optomagnonics. Mr. Muhammad Umar Siddiqui (Assistant Professor): MS from BUIITEMS, Pakistan.

Research focus: Semiconductor physics. Department of Mathematical Sciences

Overview, Specializations, and Future Extensions

Established in 2003, the Department of Mathematical Sciences stands as one of the largest academic units within BUIITEMS. Beyond running its own undergraduate and graduate degree tracks, the department delivers foundational mathematics coursework for all other departments across the university. The curriculum aligns with HEC guidelines, offering specializations across pure, applied, computational, and financial mathematics. Faculty and students actively participate in regional,

national, and international scientific conferences. The department's research interests span both abstract and applied mathematics, and it is on track to launch its official PhD program. To keep students updated on events and academic opportunities, the department maintains a dedicated digital outreach channel on Facebook under the name "BUIITEMS Mathematical Society." Strategic Vision and Mission

Vision: To deliver a broad and rigorous program of study that supports the core mission of BUIITEMS, providing high-quality education that meets the needs of a diverse student body. Mission: To equip students with the quantitative reasoning, computational skills, and conceptual understanding needed for their personal and professional lives, while supporting future careers in teaching, scholarship, and professional organizations.

Objectives and Scope: A degree in mathematics offers strong career versatility due to its close overlap with major engineering, medical, aeronautical, and electronic disciplines. Quantity analysts and mathematical specialists are highly sought after across almost every scientific and corporate field. BS Mathematics Semester-by-Semester Course Roadmap

First Semester: Calculus I, Elements of Set Theory and Mathematical Logic, Islamic Education/Ethics, Fundamentals of Computers, Reading and Writing Skills, Physics-I. Second Semester: Calculus II, Software Packages (Introduction to MATLAB and Maple), Introduction to Statistics, Communication and Presentation Skills, Pakistan Studies, Physics-II. Third Semester: Algebra I (Group Theory), Calculus III, Technical Report Writing and

Presentation Skills, Discrete Mathematics, Introduction to Sociology. Fourth Semester: Affine and Euclidean Geometry, Numerical Methods and Linear Programming, Linear Algebra, Algebra II (Ring Theory), Introduction to Economics. Fifth Semester: Point Set Topology, Classical Mechanics, Complex Analysis, Real Analysis I, Vector and Tensor Analysis. Sixth Semester: Differential Geometry, Mathematical Methods, Numerical Analysis, Ordinary Differential Equations, Real Analysis II.

Seventh Semester: Number Theory, Functional Analysis, Elective-1, Elective-2, Partial Differential Equations, Project-I. Eighth Semester: Probability Theory, Integral Equations, Elective-3, Elective-4, Project-II. MS and PhD Mathematics Course Layouts

MS Mathematics (First Semester): Core-I, Core-II, Core-III, Core-IV. MS Mathematics (Second Semester): Elective I, Elective II, Elective III, Elective IV. MS Mathematics (Third & Fourth Semesters): Independent Research Thesis execution. PhD Mathematics (First Semester): Elective I, Elective II, Elective III. PhD Mathematics (Second Semester): Elective IV, Elective V, Elective VI. PhD Mathematics (Third Semester)

onward): Doctoral Dissertation execution.

Pedagogical Constraint: PhD scholars are strictly barred from enrolling in courses they have already completed during their prior MS or M.Phil classes. Mathematical Sciences Faculty Profiles

Dr. Hasrat Hussain Shah (Associate Professor & Chairperson): PhD from the University of Science and Technology of China (USTC). HEC Approved Supervisor.

Research focus: General relativity and modified gravity theories. Prof. Dr. Muhammad Zubair Khan (Professor of Statistics & Director PRC BUIITEMS): PhD completed under a split scholarship pathway from Hajvery University and Aksaray University. HEC Approved Supervisor. Research focus: Applied statistics, econometrics, bio-statistics, panel data regression analysis, and financial statistics. Dr. Farooq Hussain (Associate Professor): PhD from the International Islamic

University, Islamabad. Research focus: Multiphase flows, peristalsis, viscous flows, heat transfer, nanofluid flows, and cavitation & bubble dynamics. Dr. Sardar Muhammad Hussain (Assistant Professor): PhD from the University of Sao Paulo, Brazil. Research focus: Fluid mechanics. Mr. Asghar Ali (Assistant Professor): M.Phil. from Newcastle University, UK. Research focus: M-theory and string theory. Mr. Malik Hafizullah (Assistant Professor - On Study Leave): PhD in progress at Sindh

University. Research focus: Partial Differential Equations (PDEs). Ms. Sahar Raza (Assistant Professor): MS from Air University, Islamabad. Research focus: Fluid dynamics and mathematical biology. Dr. Naheeda Iftikhar (Assistant Professor): PhD from the University of Balochistan, Quetta. Research focus: Fluid mechanics, numerical computation, and differential equations.

Dr. Hassan Shah (Assistant Professor): PhD from the University of Peshawar, Pakistan. HEC Approved Supervisor. Research focus: Gravitational collapse of massive objects, black hole thermodynamics, and the propagation of gravitational waves. Mr. Irfan Ali (Assistant Professor - On Study Leave): PhD in progress at Ghent University, Belgium. Research focus: Geometric function theory, non-linear analysis, and harmonic analysis. Mr. Qaisar Khan (Assistant Professor - On Study Leave): PhD in progress. Research

focus: Group theory (coset diagrams), history of mathematics, and mathematics education. Mr. Attaullah (Assistant Professor): PhD in progress in the USA. Research focus: Computational fluid dynamics. Dr. Abdul Quayam Khan (Lecturer): PhD from the Lahore University of Management

Sciences (LUMS). Ms. Ayesha Saif (Lecturer): MS from BUIITEMS, Quetta. Research focus: Special functions. Mr. Najeebullah (Lecturer): MS from BUIITEMS, Quetta. Research focus: Geometric function theory. Ms. Maryam Safdar (Lecturer): MS from BUIITEMS, Quetta. Research focus: Special

functions. Mr. Mudassar Iqbal (Lecturer - On Study Leave): PhD in progress at Universiti Teknologi PETRONAS (UTP), Malaysia. Research focus: Computer graphics, geometric curves and surfaces, numerical methods, computational algebra, surface modeling, and visually pleasing graphical models. Mr. Abdul Wahid (Lecturer): MS from BUIITEMS, Quetta. Research focus: Mathematical

biology and mathematical modeling. Mr. Aziz ur Rehman (Lecturer): MS from BUIITEMS, Quetta. Research focus: Fluid mechanics. Mr. Hazrat Ali (Lecturer): M.Phil. from Hazara University, Mansehra. Research focus: Numerical techniques and solutions for systems of ODEs and PDEs. Ms. Faiza (Lecturer): MS from the National University of Sciences and Technology (NUST), Islamabad. Research focus: Fractional

difference equations.

Ms. Farah Naz (Lecturer): MSc from COMSATS, Islamabad. Research focus: Fuzzy set theory. Mr. Bakhtiar Ahmed (Lecturer): M.Phil. from Quaid-i-Azam University, Islamabad. Research focus: Computational group theory and the theory of D-modules. Mr. Maqbool Ahmed (Lecturer in Statistics): M.Phil. from LUAWMS. Research focus: Structural equation modeling, statistical machine learning and its applications, statistical

genetics, and computational biology. Ms. Arifa Jahangir (Lecturer in Statistics): MS from the International Islamic University, Islamabad. Research focus: Mixture models and applied statistics. Ms. Saima Fareed (Lecturer in Statistics): MSc from Gomal University, Dera Ismail Khan. Research focus: Financial time series analysis. Mr. Manzoor Ahmed (Lecturer in Statistics): M.Phil. in Econometrics from the Pakistan Institute of Development Economics (PIDE), Islamabad. Research focus: Financial time

series analysis, machine learning, and structural equation modeling (SEM). Mr. Muhammad Bilal (Lecturer in Statistics - On Study Leave): PhD in progress at Abdul Wali Khan University, Mardan. Research focus: Applied statistics. Ms. Faiza Saeed (Lecturer in Statistics): MS from BUIITEMS. Research focus: Environmental statistics. Ms. Farida Aslam (Lecturer): MS from SBK Women's University, Quetta. Research

focus: Fluid dynamics. Mr. Syed Abdul Rehman (Lecturer in Statistics - On Study Leave): PhD in progress at Quaid-i-Azam University (QAU), Islamabad. Research focus: Statistics. Mr. Tahir Mehmood (Lecturer - On Study Leave): PhD in progress at SMS, Lahore. Research focus: Optimization, linear programming, and differential equations.

Mr. Mohammad Yaseen (Lecturer - On Study Leave): PhD in progress at the University of Balochistan, Quetta. Research focus: Optimization, mathematical modeling, and differential equations.